Shyam Banerjee

Bachelors in Statistics
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Course	College/University	Year	CGPA/%
Bachelors in Statistics	Indian Statistical Institute	2024-Present	82.0
Intermediate/+2	Gitanjali Public School	2024	94.8
High School	Gitanjali Public School	2022	97.4

SCHOLASTIC ACHIEVEMENTS

- Secured **rank 9** in Madhava Mathematics Competition [2026]
- Secured **rank 48** in Simon Marais Mathematics Competition [2026]
- Finalist at Trilytics Case Analytics (IIM Calcutta) [2025]
- Cleared IOQM (Regional) [2022]

WORK EXPERIENCE

- **Software Engineer Intern | Alumnus Software Ltd.** [Apr'26-Sep'26]
  - Explored convergence patterns in anomaly event sequences across PCB production systems
  - Found evidence of clusterable operating regimes in manufacturing-cycle data
- **Summer School in Astrostatistics | Penn State University** [Jun'25]
  - Participated in an intensive program on Bayesian methods, statistical inference, and astronomical data analysis.

RESEARCH EXPERIENCE

- **Posterior Inference and Gaussian Process Modeling of Stellar Variability | *Independent Research*** [May'26-Present]
  - Produced a 19-page research manuscript and open-source implementation.
  - Developed a simulation framework for periodic stellar signals corrupted by Gaussian Process noise.
  - Implemented GP-based probabilistic imputation and uncertainty quantification on Kepler stellar light curves.
  - Performed Bayesian posterior inference for hidden periodic structure using harmonic signal models.
  - Investigated inference behaviour across quiet and rotationally variable stars using real Kepler observations.

PROJECTS

- **Non-stationary Time Series Modeling | Dr. Shyamal Krishna De** [Apr'26-May'26]
  - Developed a seasonal ARIMA forecasting pipeline for weekly PM2.5 concentration data from CPCB Delhi using $\sim$8 years of observations.
  - Performed nonstationarity diagnosis using ADF, KPSS, Mann–Kendall, and Seasonal Mann–Kendall tests, followed by Box–Cox variance stabilisation and iterative seasonal/regular differencing.
  - Implemented forecasting and residual diagnostics in `R` using `forecast`, `tseries`, and `urca`; evaluated probabilistic forecast coverage on held-out test data.

SKILLS & INTERESTS

- **Programming Languages**: Python, C++, R
- **Tools & Libraries**: NumPy, SciPy, Pandas, Matplotlib, Streamlit, Scikit-learn, LaTeX
- **Interests**: Time Series, Astrostats, ML, Quantitative Finance

RELEVANT COURSES

- **Statistical Inference | Linear Statistical Models | Probability Theory | Stochastic Process | Statistical Methods | Linear Algebra | Design and Analysis of Algorithms | Differential Equations | Sample Survey** [B.Stat]

VOLUNTEERING EXPERIENCE

- **Mentoring | AP Tutors** [Sep'25-Mar'26]
  - Guided students for AP Stats and SAT Maths

EXTRACURRICULAR ACTIVITIES

- Competitive Programming (Codeforces)
- Game Theoretic Puzzles